A COMPREHENSIVE LITERATURE REVIEW AND FUTURE RESEARCH OPPORTUNITIES IN OPTIMIZATION APPROACHES IN PROJECT PORTFOLIO MANAGEMENT UNDER UNCERTAINTY AND IN DYNAMIC ENVIRONMENTS

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1. Abstract

This paper presents a comprehensive literature review of various methodologies in project portfolio management under uncertainty and resource constraints. By analyzing eight key studies, the review explores approaches such as multistage stochastic programming, risk-averse budget allocation, and project scheduling with multi-skilled human resources. The findings highlight the similarities, differences, and strengths of these models, along with their practical applications across sectors like healthcare, modular construction, hurricane emergency response, and Research & Development (R&D). The review also identifies critical gaps in the current body of research and proposes future directions for enhancing the models' adaptability, flexibility, and multi-objective optimization capabilities. These insights lay a foundation for developing more robust project portfolio management strategies in dynamic environments in order to optimize budget, time, and resources. Future research should focus on dynamic models that prioritize risk-based decision-making, enabling real-time resource reallocation and optimization.

1. Introduction

Project portfolio management (PPM) faces increasing complexity due to uncertainty in resource availability, fluctuating demands, and evolving project scopes. Traditional models, like the Triple Constraint paradigm, often fail to handle dynamic environments effectively. *The Mindset for Creating Project Value* critiques this rigidity, advocating for a shift toward entrepreneurial thinking that leverages uncertainties as opportunities rather than risks to be mitigated [11]. Similarly, *Risk-Averse Optimization and Control* emphasizes that effective PPM should integrate dynamic decision-making based on project risks, enabling continuous resource adaptation as new information becomes available [7]. This review aims to synthesize existing research, compare different methodologies, and suggest future directions for models that integrate both dynamic resource allocation and risk-based decision-making, fostering more adaptable and resilient project management frameworks.

2. Literature Review/Background

In this study, I reviewed eight key papers from various journals to understand the different approaches and methodologies in project portfolio management. The papers reviewed include:

i. "A Multistage Stochastic Programming Approach in Project Selection and Scheduling" (International Journal of Advanced Manufacturing Technology).
ii. "Stochastic Project Management: Multiple Projects with Multi-skilled Human Resources" (Journal of Scheduling).
iii. "Robust and Resilience Budget Allocation for Projects with a Risk-averse Approach: A Case Study in Healthcare Projects" (Computers & Industrial Engineering).
iv. "Risk-averse Supply Chain for Modular Construction Projects" (Automation in Construction).
v. " A clustering-based review on project portfolio optimization methods (*International Transactions in Operational Research).*
vi. "Integrated Framework for Project Portfolio Planning Management Under Resource Constraints" (Master's thesis, Stevens Institute of Technology).
vii. " Robust optimization for hurricane preparedness (*International Journal of Production Economics).*
viii. "Optimization of renewable energy project portfolio selection using hybrid AIS-AFS algorithm in an international case study (*Scientific Reports*).

3. Literature Review

3.A. Multistage Stochastic Programming Approach in Project Selection and Scheduling (International Journal of Advanced Manufacturing Technology)

This paper uses a multistage stochastic programming approach to explore the combined challenges of project selection and scheduling under uncertain conditions. The model aims to assist decision-makers in selecting projects and scheduling tasks while managing resource constraints and uncertainty in factors like costs, revenues, and resource availability [1]. The primary objective is to maximize the present worth of total profit from the chosen projects [1].

The study introduces a multi-period project selection and scheduling problem, effectively modeled by multistage stochastic programs, which are well-suited for long-term planning in uncertain environments. To capture this uncertainty, a set of scenarios with corresponding probabilities is utilized, representing the multivariate random data processes (e.g., costs, revenues, available budgets, and success chances) [1]. Given the computational complexity involved, a scenario tree is constructed using scenario reduction algorithms to streamline the analysis [1].

The model assumes that project resources are limited and renewable, focusing on maximizing project profitability. A case study is presented to validate the approach, showcasing the model's effectiveness through numerical results [1]. The scenario tree method models uncertainty, while the scenario reduction algorithm reduces computational complexity by narrowing down the number of possible scenarios [1]. This enables optimal decision-making across multiple stages, considering task precedence and resource constraints. The study's findings demonstrate the potential of multistage stochastic programming for enhancing decision-making in uncertain project environments, offering a robust solution for managing complex project portfolios.

3.A.1. Key Components of the Model

The authors use a stochastic optimization framework that includes objective functions for maximizing present value of total profit and minimizing costs, subject to resource constraints [1]. The methodology involves solving a series of interconnected linear programming problems that represent different stages and scenarios [1]. The key components of each of each model included the following:

1. Objective: Maximize the expected present value of the profit by selecting and scheduling tasks over multiple periods.
2. Scenario Tree: A set of possible future outcomes (scenarios) is generated, based on random variables such as resource availability. Scenario reduction is used to reduce the computational complexity.
3. Multistage Stochastic Programming: This allows decision-making under uncertainty over multiple time periods, where decisions are based on known information at each stage without anticipating future unknown outcomes.
4. Non-Anticipativity: Ensures that decisions at any time period depend only on current and past information, not on future events.
5. Constraints: Ensure that tasks follow a logical sequence (task precedence) and respect resource limits in each time period.

3.A.2. Multistage Linear Programming Model

The paper first delves into the linear structure of multistage stochastic programming problems, emphasizing their application in project portfolio selection and scheduling. These models utilize a set of scenarios and corresponding probabilities to model multivariate random data processes such as costs, revenues, available budgets, and the chance of success [1]. Given the computational complexity and time limitations of solving optimization problems that include all possible scenarios, scenario reduction algorithms are introduced to select a smaller, manageable subset of scenarios while preserving the essence of the original problem [1].

The general linear programming formulation in the multistage context is described as Equation 1 below:

Equation 1: Multi Linear Programming Model

Objective Function:

$$\mathbf{Minimize}\ \mathrm{C^T X}$$

Subject to:

$$\mathrm{A_1 x_1 = b_1}$$
$$\mathrm{B_i x_i + A_i x_i = b_i}$$
$$\mathrm{x_i} \in (0,1)^{\mathrm{n}} \text{ for } \mathrm{i = 1, \dots, T}$$

Objective: Minimize the total cost, $\mathrm{C^T X}$, where C is the cost vector and X is the decision matrix[1].
Constraints:

- $\mathrm{A_1 x_1 = b_1}$: Initial condition ensuring that the first stage respects the resources available.
- $\mathrm{B_i x_i + A_i x_i = b_i}$: for every subsequent stage i, the decision $\mathrm{x_i}$ depends on the previous stage's decisions and the resource availability.
- $\mathrm{x_i} \in (0,1)^{\mathrm{n}}$: The decision variables are binary, indicating a task is selected (1) or not selected (0).
- $C = [C_1,\ C_2, \dots, C_n]^T$ and $C_i$is an n-dimensional vector, for i=1,2,…, T.
- $C_1, b_1, and\ A_1$ are known, but some (or all) of the entries of the cost vectors C_t, matrices B_t , A_t, and right hand side vectors $b_1, t = 2, \dots, T$, are random.

This equation forms the core of the multistage programming model, where decisions are made sequentially (i.e. Decision($\mathrm{x_1}$), Observation ξ_2: $C_2,\ B_2, A_2,\ b_2$); $\mathrm{Decision(x_2)}, \dots, \mathrm{Observation}\ (\xi_T$: C_T, $B_T, A_T,\ b_{2T}$, $\mathrm{Decision(x_T)}$ [1].

3.A.3. Scenario Tree Model

This formulation extends the previous linear programming model to handle the multiple uncertain future scenarios.
Equation 2: Senario Tree Model

Objective Function:

$$Minimize \sum_{k=1}^{K} p_k C_k^T x_k$$

Subject to:

$$A_1 x_1^k = b_1$$
$$B_i^k x_{i-1}^k + A_i^k x_i^k = b_i^k, i = 2, \dots, T$$
$$x_i^k \in \{0,1\}^n, \forall i = 1, \dots, T, \forall k = 1, \dots, K$$

The objective is to minimize the expected total cost over all scenarios. Here, p_k is the probability of scenario *k*, and C_k is the cost vector for scenario *k* [1].

Scenario-specific Constraints:

- $A_1 x_1^k = b_1$: The initial condition in each scenario.
- $B_i^k x_{i-1}^k + A_i^k x_i^k$: The constraints depend on scenario *k*, where each scenario may have different parameters for costs and resources.
- $x_i^k \in \{0,1\}^n$: Binary decision variables in each scenario and time period.

In the article, the authors note that Equation 2 is "tedious and impractical" and later modify this equation into the Scenario Tree Reduction Algorithm to reduce the complexity of the problem [1].

3.A.4. Multistage Stochastic Programming Model

From a linear programming model and then applying concepts of scenario trees, the authors evolved this concept into the Multistage Stochastic Programming model below [1]. The model uses a scenario tree to represent different realizations of uncertain parameters (such as project outcomes, profits, and resource availability) over multiple periods. The model's primary objective is to maximize the expected total profit while ensuring that tasks are scheduled correctly, and resources are optimally allocated [1].

Equation 3: Multistage Stochastic Programming Model

Objective Function:

$$\text{Maximize} \sum_{n \in N} \pi_n \sum_{i=1}^{I} \sum_{j=1}^{J_i} P_{ij}^n X_{ij}^n$$

Objective: The objective is to maximize the total expected profit across all projects, tasks, and scenarios. Each term in the equation contributes to the overall expected value by accounting for different scenarios.

The summation over $n \in N$, *n* represents a node in a scenario tree, which is part of a set of scenarios *N.* Scenarios model uncertainty (e.g., variations in cost, revenue, or resource availability), and each node *n* represents a possible outcome at a given stage in the scenario tree.

π_n is the probability associated with scenario *n*. It represents the likelihood of that scenario occurring. The sum of all π_n values equals 1, ensuring the probabilities are valid.

The summation over *i* (Projects), i indexes the set of projects *i*=1,2,…,*I*. The total profit is computed for each project separately, and then combined for all projects.

The summation over *j* (Tasks), **i** indexes the set of projects *i*=1,2,…,*I*. The total profit is computed for each project separately, and then combined for all projects.

P_{ij}^n This represents the profit (or present value) of task *j* from project *i* in scenario *n*. Since the profit depends on the scenario, it is indexed by *n*.

X_{ij}^n is a binary decision variable (0 or 1) indicating whether task *j* in project *i* is selected in scenario n. If $X_{ij}^n = 1$, task *j* is started; if $X_{ij}^n = 0$, it is not started.

This objective function has four constraints (e.g. Task Precedence Constraint, Resource Constraint, Task Selection Constraint, and Binary Constraint. I will explain these further below.

Task Precedence Constraint is a constraint that ensures tasks are scheduled in the correct order, respecting dependencies between tasks [1]. This ensures that task j+1j+1 can only start after task *j* is completed. The equation is as follows:

Equation 4: Task Precedence Constraint

$$\mathrm{d}_{ij} X_{i(j+1)}^{n} \quad \leq \sum_{\gamma=1}^{t(n)-1} (t(n) - l) X_{ij}^{n-\gamma}; l < (n), \forall n \in N$$

Here, d_{ij} represents the minimum time lag between tasks *j* and *j*+1 due to task duration [1].

The Resource Constraint is the constraint that ensures that the total resource consumption across all tasks in any time period t does not exceed the available resources [1]. The equation is as follows:

Equation 5: Resource Constraint

$$\sum_{i=0}^{\mathrm{I}} \sum_{j=0}^{J_i} r_{ijk}^{n} \sum_{v=0}^{d_{ij}-1} \quad \mathrm{X}_{\mathrm{ij}}^{\mathrm{n-v}} \leq R_k^n \ \forall k = 1, \ldots, K, n \in N$$

Where $r_{ijk}^{n}$is the resource requirement of task *j* in project *I* for resource *k* under scenario *n*, and R_k^n is the available amount of resource *k* in scenario *n* [1]. Additionally, the Binary Decision Variables $\mathrm{X}_{\mathrm{ij}}^{\mathrm{n}} \in \{0,1\}$ indicates whether or not a task *j* in project *i* is selected (1) or not (0) in scenario *n* [1].

The task selection constraint is when each task can be selected at most once in the scheduling horizon. This prevents the task from being scheduled more than once [1].

Equation 6: Task Selection Constraint

$$\sum_{l=1}^{t(n)} X_{ij}^{n-\gamma} \leq 1;\ \forall i, \forall j = 1, \ldots, J_i, \forall n \in N$$

The binary constraint is where decision variables X_{ij}^{n}must be binary (either the task is selected, or it is not).

Equation 7: Binary Constraint

$$X_{ij}^{n} \in \{0,1\}, \forall i = 1 \ldots, I, \forall j = 1, \ldots, J_i, n \in N$$

This equation seeks to maximize the expected total profit from selecting and scheduling tasks across multiple projects and scenarios. It does so by considering the probability-weighted profits in different scenarios and ensuring the decision variables X_{ij}^{n} (representing task selections) are optimized [1]. The model is subject to constraints ensuring task precedence, resource availability, and single-task execution per project [1].

3.A.5. Case Study

The case study in the paper explores the application of a multistage stochastic programming model for project selection and scheduling under uncertainty, focusing on a Research & Development (R&D) company [1]. The company needs to schedule two R&D projects, each consisting of multiple tasks, over 12 months [1]. The main objective is to maximize the total profit by selecting and scheduling tasks efficiently, considering the uncertain availability of financial and human resources [1]. Using

Equation 3: Multistage Stochastic Programming Model, the model solved the problem within 15 CPU time units using MATLAB, achieving an optimal profit of 107 units [1]. This reflects the present value of profits from the two R&D projects over 12 months. The tasks were scheduled in a way that maximized profit while respecting the task precedence and resource availability constraints [1]. The scenario tree model captures the uncertainty in resources, allowing the company to make decisions that balance risk and profit [1]. By reducing the scenario tree from 100 scenarios with 2,400 nodes to 84 scenarios, the overall computational complexity was lowered while retaining the accuracy of the model [1].

3.A.6. Gaps in Study and Opportunities for Future Research

While the case study demonstrates the power of stochastic programming for project scheduling under uncertainty, there are numerous opportunities for improving the model by incorporating more complex, realistic constraints and objectives. Future research could focus on making the model more adaptive, flexible, and capable of handling multiple objectives and dynamic changes.

Table 1: Gaps in Study and Opportunities for Future Research below covers the gaps and opportunities for future research.

Table 1: Gaps in Study and Opportunities for Future Research

Identified Gaps In Study	Description	Opportunities for Future Research
Expansion into Non-Financial Parameters	The study focuses primarily on resource availability in terms of financial and human resources. Future research could consider other non-financial uncertainties, such as technology changes, market demands, or project interdependencies.	Incorporating non-financial parameters like risk assessment, technology advancements, and competitor actions would provide a more comprehensive decision-making framework.
Incorporation of Time-Dependent Resource Constraint	The current model assumes that resources (budget, staff) are available at specific points in time and do not change dynamically during the project. However, real-world projects often face time-dependent variations in resource availability, such as sudden budget cuts or shifts in personnel.	Future models could introduce dynamic resource constraints, where resource availability fluctuates within a project based on external factors (e.g., company funding cycles or staff attrition).
Multi-Objective Optimization	The study focuses on a single objective: maximizing profit. However, many real-world scenarios require a multi-objective approach, balancing not just profit but also factors like project completion time, risk minimization, and social impact.	Future research could explore multi-objective stochastic programming, where objectives like profit, project completion time, and risk mitigation are considered simultaneously. This could provide a more robust decision-making tool for complex projects with competing goals.

Project Cancelations and Flexibility	The current model does not account for the possibility of cancelling or delaying projects due to unfavorable scenarios. In practice, organizations often need the flexibility to cancel projects or pause tasks when resources are severely constrained.	Introducing flexibility in project selection, allowing for the cancellation or rescheduling of projects in response to scenario outcomes, would provide more realistic solutions.
Real-Time Decision Making	The model operates in a batch mode, where all decisions are made at the beginning of the planning period. However, in many real-world cases, decisions must be updated in real-time as new information becomes available (e.g., new budget updates or unforeseen delays).	Developing real-time stochastic decision-making models that allow for continuous updates and adaptive scheduling as new data arrives would improve the practicality of the model.

While the case study demonstrates the power of stochastic programming for project scheduling under uncertainty, there are numerous opportunities for improving the model by incorporating more complex, realistic constraints and objectives. Future research could focus on making the model more adaptive, flexible, and capable of handling multiple objectives and dynamic changes.

3.B. "Stochastic Project Management: Multiple Projects with Multi-skilled Human Resources" (Journal of Scheduling).

The article titled "Stochastic Project Management: Multiple Projects with Multi-Skilled Human Resources"presents two stochastic optimization approaches for solving project scheduling and personnel staffing problems under uncertainty [2]. It extends a deterministic model developed by Heimerl and Kolisch by introducing stochasticity in work package processing times [2]. The authors focus on the problem of scheduling multiple projects that share human resources with heterogeneous skill levels. The main challenge is that work time demand is uncertain, requiring an optimal balance between internal resources and the more costly external resources when demand exceeds internal capacity [2].

The two solution approaches discussed are:

1. Matheuristic Approach: This combines a metaheuristic for scheduling projects with the Frank-Wolfe algorithm for optimizing the staffing subproblem [2]. The metaheuristic uses Iterated Local Search (ILS) with variable neighborhood descent to explore potential solutions [2].
2. Sample Average Approximation (SAA): This method approximates the stochastic problem by solving multiple deterministic instances using mixed-integer programming based on sampled scenarios [2].

The primary goal of the study is to minimize the expected cost of using external human resources [2]. The model is tested on synthetically generated instances inspired by real-world data from an IT department in a large semiconductor company [2]. Results show that the metaheuristic approach provides robust performance, particularly for small and medium-sized problem instances, while the SAA performs well under certain conditions but struggles with larger instances [2].

3.B.1. Objective Function in the Metheuristic and SAA Approaches

Both approaches share some foundational equations, but there are important distinctions in how they handle the stochastic optimization problem for project scheduling and staffing [2].
For a project *p* and a time period t, where the authors say:

Equation 8:Objective Function in Matheuristic and SSA Approaches

$$\tau_{pt} = \{q \in \{1, \ldots, d_p\} | ES_{pq} \leq t \leq LS_{pq}\}$$

Objective Function:

$$Minimize \sum_{p \in P} \sum_{t=ES_p}^{LF_p} \sum_{s \in S} c_s^e \, \mathrm{E}\left(\left[\sum_{q \in \tau_{pt}} D_{psq} z_{pqt} - \sum_{k \epsilon K_s} x_{ptsk}\right]^+\right)$$

Where:

- *P*: Set of projects.
- *t*: Time period from earliest start ES_p to latest finish LF_p for project *p*.
- *s*: Skill required for a project.
- c_s^e: Cost per unit of external resource for skill *s*.
- D_{psq}: Random variable representing the demand of work package *q* for project *p* requiring skill *s*.
- z_{pqt}: Binary variable that equals 1 if activity *q* of project *p* is scheduled in period *t*, 0 otherwise.
- x_{ptsk}: Amount of work done by internal resource *k* with skill *s* for project *p* in time period *t*.
- E: Expectation operator.
- E([·])+: E denotes the expectation and x^+ Indicates the non-negative portion of the difference (maximization of 0 and the expression inside).
- Q represents an activity of project p in a predefined time window of Early Start (ES_{pq}) and Late Start (LS_{pq}).

This equation aims to minimize the expected cost of external costs[2]. External costs are used when the demand for a particular skill in a project exceeds the capacity of internally scheduled capacity or resources [2].

3.B.1.1. Activity Scheduling Constraints

The following constraints ensure that the project tasks are scheduled correctly.

Equation 9: Activity Scheduling Constraints

$$\sum_{t=ES_{pq}}^{LS_{pq}} z_{pqt} = 1 \; \forall p \in P, q = 1, \ldots, d_p$$

This ensures that each activity *q* of project *p* is scheduled exactly once within its predefined time window [ES_{pq},LS_{pq}]. The binary variable z_{pqt} determines when each activity is scheduled.

3.B.1.2. Precedence Constraint

This constraint ensures that the activities within a project are scheduled in the correct order. The finish time of one activity must be before the start time of the next activity in the sequence.

Equation 10: Precedence Constraint

$$\sum_{t=ES_{pq}}^{LS_{pq}} t z_{pqt} < \sum_{t=ES_{p,q+1}}^{LS_{p,q+1}} t z_{p,q+1,t} \; \forall p \in P, q = 1, \ldots, d_p - 1,$$

3.B.1.3. Resource Capacity Constraint

The resource capacity constraints ensure that internal resources are not over-utilized.

Equation 11: Resource Capacity Constraint

$\sum_{p \in P} \sum_{s \in S} \frac{1}{n_{sk}} x_{ptsk} \leq a_{kt} \; \forall k \in K, t = 1, \dots, T, p \in P$

This constraint ensures that the total amount of work assigned to each internal resource *k* in period *t* does not exceed the available capacity a_{kt} for that time. The internal resource's efficiency n_{sk} (how well resource *k* performs skill *s*) is used to calculate the actual time required to complete a work package. The higher the efficiency, the less time is needed.

3.B.1.4. Decision Variable

The following define the decision variables for scheduling and staffing.

Equation 12: Non-Negativity Constraints

$$x_{ptsk} \geq 0 \; \forall p \in P, t = ES_p, \dots, LF_p, s \in S, k \in K_s$$

The decision variable x_{ptsk} , which represents the amount of work done by internal resource *k* for skill *s* in project *p*, must be non-negative [2].

Equation 13: Binary Decision Variables for Scheduling

$$z_{pqt} \in \{0,1\} \; \forall p \in P, q = 1, \dots, d_p, t = ES_{pq}, \dots, LF_{pq}$$

The binary decision variable z_{pqt} determines whether activity *q* of project *p* is scheduled in period *t* (1 if scheduled, 0 if not) [2].

3.B.2. Equations in the Matheuristic Approach

The matheuristic approach solves the problem by breaking it down into two subproblems: project scheduling and staffing. The project scheduling part is handled by the iterated local search (ILS) metaheuristic, while the staffing problem is solved using the Frank-Wolfe algorithm for convex optimization [2].

Equation 14: Staffing Subproblem (Convex Optimization using Frank-Wolfe Algorithm)

$$Minimize \sum_{p \in P} \sum_{t=ES_p}^{LF_p} \sum_{s \in S} c_s^e \, \mathrm{E}\left(\left[\sum_{q \in \tau_{pt}} D'_{psq} - \sum_{k \in K_s} x_{ptsk} \right]^+ \right)$$

For a fixed project schedule *z*, this subproblem minimizes the external cost of staffing by optimizing the allocation of internal resources. The demand D'_{psq} for each project *p*, skill *s*, and period *t* is treated as a random variable, and internal resources are allocated based on their efficiency. If internal resources cannot meet the demand, external resources are used, and the objective is to minimize the expected external cost [2].

At each iteration, the Frank-Wolfe algorithm solves a linear approximation of the staffing problem. It minimizes the convex cost function by iteratively improving the resource allocation and reducing external costs [2].

3.B.3. Equations in the Sample Average Approximation (SAA) Approach

In the Sample Average Approximation (SAA) approach, the stochastic problem is approximated by generating several sample scenarios of the random variables (i.e., different realizations of work package demands). The expected cost is then approximated by averaging over these scenarios.

For each sample scenario where $n \in N$, the following deterministic equivalent problem is solved.
Equation 15: Deterministic Equivalent Problem

$$Minimize\ \frac{1}{N}\sum_{n=1}\sum_{p\in P}\sum_{t=ES_p}^{LF_p}\sum_{s\in S} c_s^e y_{pts}^{(n)}$$

The external labor cost $y_{pts}^{(n)}$for project *p*, skill *s*, and period *t* in scenario *n* is minimized across all sample scenarios. The objective is to minimize the average cost over all scenarios. The deterministic equivalent problem is a mixed-integer linear program (MILP) that can be solved using solvers.

Both approaches share the same foundational equations (objective function, scheduling constraints, and resource capacity constraints), but they differ in how they solve the problem. The matheuristic approach uses an iterative local search and convex optimization, while SAA relies on scenario-based sampling and deterministic optimization.

3.B.4. Experimental Results

Overall, the Matheurist model experimental results performed better for small and medium-sized test instances. The SAA model failed to return any results. The results also proved that some managerial insights are valid regarding uncertainty on required efforts.

3.B.5. Gaps in Study and Opportunities for Future Research

While the study successfully extends prior deterministic models of project scheduling and staffing to a stochastic setting, several limitations and opportunities for improvement remain, especially time-dependent constraints and risk management. The current model provides valuable insights into resource allocation under uncertainty, but it assumes static scheduling and risk-neutral optimization, which do not fully capture the complexities of real-world projects. Projects often face dynamic changes over time, and decisions must adapt to fluctuating risk profiles, particularly for larger, high-stakes projects. In this section, I will highlight key gaps in the study, emphasizing the role of time and risk in project scheduling and staffing decisions. I will also propose areas for future research that explore how time-dependent factors and risk aversion can improve project outcomes and better reflect the challenges faced in complex, uncertain environments. Table 2: Stochastic Project Management: Multiple Projects with Multi-Skilled Human Resources Journal Gaps in Study and Opportunities for Research below shows all of the gaps and future research opportunities.

Table 2: Stochastic Project Management: Multiple Projects with Multi-Skilled Human Resources Journal Gaps in Study and Opportunities for Research

Identified Gaps In Study	Description	Opportunities for Future Research
Limited Consideration of Resource Substitutability	The study focuses primarily on the substitutability of internal resources with external resources in cases of insufficient internal capacity. However, it does not explore within-skill or between-skill substitutions of internal resources by other internal resources, which is a common recourse action during work overload. This limitation suggests a gap in addressing how organizations can optimize internal resource flexibility to reduce external costs further.	Future research should develop models that incorporate these forms of reactive planning and explore solution techniques to handle the complexity of this extended framework.
Risk-Neutral Assumptions	The current model operates under the assumption of risk neutrality, which is appropriate for multi-project management with relatively small or medium-sized projects. However, in cases where larger projects or higher-risk scenarios are involved, risk-neutral optimization may not be suitable.	Future studies should investigate risk-averse optimization approaches, which account for specific risks associated with large, complex projects. Incorporating risk aversion would offer a more conservative and practical planning strategy in high-stakes environments.

3.C. "Robust and Resilience Budget Allocation for Projects with a Risk-averse Approach: A Case Study in Healthcare Projects" (Computers & Industrial Engineering).

This paper addresses the critical role of Budget Allocation for Projects (BAP) in completing large-scale projects, particularly in sectors like healthcare where timely execution is essential. The study introduces an innovative approach called Robust, Resilience, and Risk-Averse Budget Allocation (3RBAP), which combines robust stochastic optimization with a focus on worst-case scenarios to enhance budget allocation strategies under uncertainty. By minimizing weighted expected progress functions and incorporating minimum progress functions, this model aims to effectively manage budget fluctuations and improve project outcomes in healthcare construction.

3.C.1. Main Equations and Parameters

The objective is to maximize progress by distributing the budget optimally across various projects while considering both expected and minimum completion rates [3].

The objective function maximizes the weighted expected and minimum the physical percent complete.

Equation 16: 3RBAP Mathematical Model

Objective Function:

$$Maximize\ Z = (1-\beta)\Gamma\Gamma + \beta\Gamma$$

Subject to:

$$\Gamma\Gamma = \sum_{s} pp_s \Gamma_s$$

Where:

- Z is the overall objective, representing optimized progress across all projects.

- β is the conservative coefficient (risk aversion), determining how much weight is given to the worst-case scenario.
- $\Gamma\Gamma$ is the expected project completion across all scenarios.
- Γ_s is the minimum completion rate across all scenarios.
- pp_s is the probability of scenario ss.

Equation 17: Progress constraints

$$\Gamma = min_s(\Gamma'_s), \qquad \forall i, s$$

$$\Gamma'_s = \frac{\sum_i ff_{is}}{|I|}, \qquad \forall s$$

$$ff_{is} = w_i kk_{is}, \qquad \forall i, s$$

Equation 18: Budget Constraints:

$$kk_{is} = \frac{x_{is}}{bac_{is}}, Type\ 1\ Linear, \forall i, s$$

Or

$$kk_{is} = (\frac{x_{is}}{bac_{is}})^{\,n}(1 - p_{is}), n \geq 0, \qquad Type\ 2\ Nonlinear\ Degree, \forall i, s$$

Or

$$kk_{is} = (\frac{e^{-x_{is}} - 1}{e^{-bac_{is}} - 1})^{\,n}(1 - p_{is}), n \geq 0, \qquad Type\ 3\ Nonlinear\ Exponential, \forall i, s$$

Or

$$kk_{is} = (log_a((a-1)(\frac{x_{is}}{bac_{is}} + 1))^{\,n}((1 - p_{is}), n \geq 0, a > 0, \qquad Type\ 4\ Nonlinear\ Logarithm, \forall i, s$$

Or

$$kk_{is} = (\sin(\frac{\pi}{2}\frac{x_{is}}{bac_{is}})^{\,n}(1 - p_{is}), n \geq 0, \qquad Type\ 5\ Nonlinear\ Sine, \forall i, s$$

$$x_{is} \leq \ bac_{is,}\ \forall i, s$$

Where:

- p_{is} is the percentage of project i completed under scenario s.

Equation 19: Resiliency with the Budget Constraint

$$\sum_i x_{is} = bb_s, \qquad \forall s$$

$$bb_s \leq \ \delta MaxBud, \qquad \forall i, s$$

Where:

- x_{is} is the budget assigned to project i under scenario s.
- bb_s is the total budget allocated for scenario *s*.
- δ is the resiliency coefficient, determining how much of the maximum budget can be allocated under uncertain conditions.
- $\delta MaxBud$ is the total available budget.

Equation 20: Decision variables

$$x_{is} \geq 0, \qquad \forall i, s$$

3.C.2. Results of the Study

The research utilizes a hybrid optimization method to allocate budgets under uncertain scenarios, applying sensitivity analysis to assess the impact of robustness, resilience, and risk-aversion on project performance. Results from the case study reveal that implementing 3RBAP reduces the progress function by 17.07% compared to standard approaches. Additionally, increasing the conservative coefficient to 50% leads to a 7.32% reduction in the progress function, with minimal impact on computation time. Conversely, decreasing the resiliency coefficient disrupts budget allocation and lowers the progress function. As the problem scale grows, computation time increases steadily while the progress function continues to decrease.

The study further explores different mathematical functions, finding that switching to exponential and sine functions improves the progress function by 41.46%. Overall, the 3RBAP approach demonstrates significant potential to enhance budget management, especially in healthcare projects, by integrating robust and risk-averse strategies that effectively address uncertainty and optimize resource allocation

3.C.3. Gaps and Opportunities in Study

While the robust, risk-averse budget allocation model presented in this study offers significant advancements in handling uncertainty and minimizing project delays, several gaps remain that limit its full applicability in broader and more complex project environments. The current approach is focused primarily on budget uncertainties and assumes static resource allocation, which may not fully reflect the dynamic nature of real-world projects. Moreover, the case study's focus on healthcare projects provides valuable insights but restricts the generalizing of the model across different sectors.

In this section, I will identify key gaps in the existing study, particularly in terms of risk modeling, dynamic reallocation of resources, and computational complexity. Furthermore, I will explore potential opportunities for future research that could expand the model's effectiveness, enhance its adaptability to different industries, and reduce computational costs while maintaining the robustness of project outcomes under uncertainty. Addressing these gaps will help to improve the resilience and flexibility of multi-project management, making it better suited for a variety of industries facing diverse and unpredictable risks.

Table 3: "Robust and Resilience Budget Allocation for Projects with Risk-Averse Approach: A Case Study in Healthcare Projects" Journal Gaps and Opportunities for Future Research

Identified Gaps In Study	**Description**	**Opportunities for Future Research**
Expanding Risk Models	The model only considers a limited number of risk factors (primarily budget uncertainty) but does not account for other critical uncertainties, such as changes in workforce availability, regulatory changes, or market conditions. A more comprehensive risk model could	Future research could include more complex risk factors, such as fluctuating resource availability, market volatility, or regulatory impacts. Incorporating a broader range of uncertainties would make the model more applicable to diverse industries and environments.

	provide a better-rounded approach to project management under uncertainty.	
Incorporating Risk-Averse Decision-Making Techniques	The current model operates under the assumption of risk neutrality, which is appropriate for multi-project management with relatively small or medium-sized projects. However, in cases where larger projects or higher-risk scenarios are involved, risk-neutral optimization may not be suitable.	The study could be expanded to explore risk-aversion techniques in greater depth, particularly in relation to decision-making frameworks. This would include exploring behavioral models that account for how decision-makers respond to uncertainty in resource allocation and project progress.
Computational Complexity	The sample average approximation (SAA) used for risk-averse optimization can become computationally expensive, especially for large-scale projects with multiple uncertainties.	Future studies should investigate metaheuristic or approximation algorithms to reduce the computational burden while maintaining solution quality.

3.D. "Risk-averse Supply Chain for Modular Construction Projects" (Automation in Construction)

The paper focuses on addressing uncertainties in the logistics of modular construction projects by using a Robust Optimization (RO) approach. Modular construction, a growing trend in the industry, involves prefabricating components off-site and transporting them for assembly [4]. The logistics behind this process are complicated by factors such as weather, transportation delays, labor productivity fluctuations, and equipment malfunctions [4]. The authors propose a mathematical model that helps decision-makers design a supply chain that mitigates these uncertainties [4]. This model optimizes the selection of warehouse locations, production schedules, transportation routes, and inventory management to minimize costs while accommodating risk aversion [4].

A case study on a school dormitory project shows that the RO model outperforms the traditional two-stage stochastic programming (2SP) model by offering cost-efficient and more resilient solutions [4]. The paper's findings highlight the effectiveness of the RO approach in reducing cost variance and ensuring stable operations despite various demand uncertainties at construction sites [4].

3.D.1. Equations Parameters, and Variables

The authors define the objective of the model as a minimization of costs incurred by manufacturing, transportation, stock-keeping (inventory) and warehouse establishment under the influence of various site demand variation factors [4]. The Objective equation is as follows:

Equation 21: Minimization of the Cost Incurred by Manufacturing Transportation, Stock-Keeping (Inventory), and Wherehouse Establishment under the Influence of Various Site Demand Variation Factors

$$Minimize\ ETC + \lambda \sum_{s \epsilon S} P_s (TC_s - ETC + 2\theta_s) + \omega \sum_{s \epsilon S} P_s \delta_s$$

Where:

Equation 22: Calculate the Fixed Variable Production Costs (FPC and VPC) for the production of all types of modular Products in the Period of Both Preparation and Construction Phases

$$FPC = MF\left(\sum_{a\epsilon A} y_a + \sum_{p\epsilon P} \hat{y}_p\right)$$

$$VPC = \sum_{a\epsilon A}\sum_{j\epsilon J} m_{aj} \cdot MV_j + \max\left(m_{aj} - \frac{MMR_j}{2}, 0\right) \cdot MV_j + \sum_{p\epsilon P}\sum_{j\epsilon J} \widehat{m}_{pj} \cdot MV_j + max(\widehat{m}_{pj} - \frac{MMR_j}{2}, 0) \cdot MV_j$$

$$\widehat{IC} = \sum_{p\epsilon P}\sum_{z\epsilon Z}\sum_{j\epsilon J} \hat{\imath}^{W}_{pzj} \cdot V_j \cdot SC^W$$

Equation 23: Computes Total Inventory Cost for Each Scenario After Construction Work Starts

$$IC_s = \sum_{a\epsilon A}\sum_{j\epsilon J} i^{W}_{aj} \cdot V_j \cdot SC^F + \sum_{a\epsilon A}\sum_{z\epsilon Z}\sum_{j\epsilon J} i^{W}_{pzj} \cdot V_j \cdot SC^W + \max(i^{W}_{saj}) \cdot V_j \cdot SC^C$$

Equation 24: Captures the Transportation Cost Between the Factory and Warehouse for the Initial Inventory Preparation Period

$$\widehat{TRC}^{\overrightarrow{FW}} = \sum_{p\epsilon P}\sum_{z\epsilon Z}\sum_{j\epsilon J} (\hat{t}^{\overrightarrow{FW}}_{pzj} / L^{\overrightarrow{FW}}_j) \cdot D^{\overrightarrow{FW}}_z \cdot TC^{\overrightarrow{FW}}$$

Equation 25: Used to Model the Transportation costs Between the Factory and the Warehouse

$$TRC^{\overrightarrow{FW}} = \sum_{a\epsilon A}\sum_{z\epsilon Z}\sum_{j\epsilon J} (\hat{t}^{\overrightarrow{FW}}_{azj} / L^{\overrightarrow{FW}}_j) \cdot D^{\overrightarrow{FW}}_z \cdot TC^{\overrightarrow{FW}}$$

Equation 26:Used to Model the Transportation Costs Between the Warehouse and Construction Sites at the Beginning of Construction Works

$$TRC^{\overrightarrow{WC}} = \sum_{a\epsilon A}\sum_{z\epsilon Z}\sum_{j\epsilon J} (\hat{t}^{\overrightarrow{FW}}_{azj} / L^{\overrightarrow{FW}}_j) \cdot (D^{\overrightarrow{FW}}_z \cdot TC^{\overrightarrow{FW}} + LP_z)$$

Equation 27: Calculates the Fixed Cost on the Construction Site in Each Scenario

$$SFC_s = SF \cdot DD_s$$

Equation 28: Evaluate the Extension Cost for Each Scenario, Incurred When Weekly Demands are Not Met, and the Site is Forced to Extend the Operations Beyond the Initially Planned Period

$$EX_s = \max(x - DD_s, 0) \cdot SF$$

Equation 29: Calculates the Warehouse Establishment Cost for Choosing One of the Candidate Locations

$$WO = \sum_{z \epsilon Z} e_z WEC_z$$

Equation 30: Computes the Proxy Measure of Model Infeasibility Which is Composed by Two Parts

$$\delta_s = \sum_{a \epsilon A} \sum_{j \epsilon J} \min(i_{saj}^C, 0) \cdot + \sum_{a \epsilon A} max \left[\sum_{z \epsilon Z} \max(i_{saj}^C, 0) - N, 0 \right]$$

Equation 31: Determines the Total Cost for Each Scenario, Defined as the Sum of All the Above Cost Terms

$$TC_s = FPC + VPC + \widehat{IC} + IC_s + \widehat{TRC}^{\overrightarrow{FW}} + TRC^{\overrightarrow{FW}} + TRC^{\overrightarrow{WC}} + SFC_s + EX_s + WO$$

Equation 32: Defines the Expected Value of the Total Costs (ETC)

$$ETC = FPC + VPC + \widehat{IC} + \widehat{TRC}^{\overrightarrow{FW}} + TRC^{\overrightarrow{FW}} + TRC^{\overrightarrow{WC}} + WO + \sum_{s \in S} P_s (IC_s + SFC_s + EX_s)$$

Where:

- j: Product types; j ∈ J, J is the set of module product types.
- s: Demand scenario; s ∈ S, S is the set of demand scenarios.
- z: Warehouse location; z ∈ Z, Z is the set of warehouses.
- a: Working weeks of the construction project; a ∈ A, A={1,2,…,LW}.
- p: Weeks for preparing initial inventory; p ∈ P, P={1,2,…,PW}

Parameters: Construction Sites:

- P_s The probability that scenario ss occurs.
- DD_s: The construction duration of scenarios s in weeks.
- LW: The longest working duration among all scenarios in weeks.
- D_{saj}: The demand of product j at the site in week aa in scenario s.
- SF: The fixed overhead at the construction site per week.
- N: The maximum number of products that can be stored on the site without stacking.
- SC^C: Storage cost per cubic meter per week at the construction site.

Parameters: Warehouse:

- V_j: The volume of product j in cubic meters.
- $WCAP_z$ Maximum storage capacity for warehouse z in cubic meters.
- SC^W: Storage cost per cubic meter per week in warehouses.
- WEC_z: The cost for establishing warehouse z.

Parameters: Factory:

- MMR_j: The maximum weekly manufacturing rate of product j.
- MF: Fixed cost per week if the factory is operating.
- MV_j: Basal unitary (variable) cost for manufacturing one product j.

- FCAP: Maximum inventory capacity in the factory in cubic meters.
- PW: The last week of the initial inventory preparation period.
- SCF: Storage cost per cubic meter per week in the factory.
- M: A large positive constant used in constraints (often referred to as "big M").

Parameters: Transportation:

- $D_z^{\overrightarrow{FW}}$: The distance between factory and warehouse z in kilometers.
- $D_z^{\overrightarrow{FWC}}$: The distance between warehouse z and site in kilometers.
- $TC^{\overrightarrow{FW}}$: Transportation cost from the factory to the warehouse per truck per kilometer.
- $TC^{\overrightarrow{WC}}$: Transportation cost from the warehouse to the site per truck per kilometer.
- $L_j^{\overrightarrow{FW}}$: The quantity of product j that can be loaded onto a single truck running on the route from factory to warehouse.
- $L_j^{\overrightarrow{WC}}$: The quantity of product j that can be loaded onto a single truck running on the route from warehouse to the site.
- LP_z The lateness penalty between the site and warehouse z.

Decision Variables:

- m_{aj}: Manufacturing quantity of product j per week at the factory in week a.
- $\widehat{m}_{pj}$: Manufacturing quantity of product j per week at the factory in week p.
- e_z: A binary variable that indicates whether a certain warehouse z is established or not.
- $\hat{\imath}_{pzj}^{W}$: The quantity of initial inventory of product j in warehouse z in week p.
- ${i_{aj}}^F$: The quantity of inventory of product j in the factory in week a.
- ${i_{azj}}^W$: The quantity of inventory of product j in warehouse z in week a.
- ${i_{saj}}^C$: The quantity of inventory of product j at the construction site in week a in scenario s.
- ${\hat{t}_{pzj}}^{\overrightarrow{FW}}$: Transportation quantity for initial inventory of product j from the factory to warehouse z in week p.
- ${t_{azj}}^{\overrightarrow{FW}}$: Transportation quantity of product j from the factory to warehouse z in week aa.
- ${t_{azj}}^{\overrightarrow{WC}}$: Transportation quantity of product j from warehouse z to the site in week aa.
- $\hat{y}_p$: A binary variable representing whether the factory is operating in week p (preparation period).
- y_a: A binary variable representing whether the factory is operating in week a (construction period).
- x: The total duration of the optimal transportation scheme in weeks.
- c_a: A binary variable that indicates whether any product is sent to the site in week a.
- θ: A dummy decision variable for linearizing the second term of the objective function.
- λ: A parameter controlling the trade-off between solution and model robustness (set to 1).
- ω: A parameter controlling the trade-off between solution and model robustness (set to 1).

3.D.2. Study Results

The study's results demonstrated the effectiveness of the proposed Robust Optimization (RO) model in managing supply chain uncertainties in modular construction projects. Applied to a school dormitory construction case study, the model optimized production schedules, transportation plans, and warehouse location selection under various demand scenarios. The RO model's strength lies in its ability to handle multiple sources of uncertainty, such as bad weather and labor disruptions, ensuring that the operational costs remain within a manageable range regardless of the scenario.

Compared to the two-stage stochastic programming (2SP) model, the RO approach provided better risk management. Although the 2SP model slightly outperformed RO in expected cost savings by 0.38%, the RO model offered substantial improvements in reducing cost variance by 58.53% and the standard deviation of operational costs by 25.91%. Additionally, the RO model minimized the risk of cost overruns in worst-case scenarios, reducing the maximum operational cost by £6380 compared to the 2SP model.

The study also identified the optimal warehouse location (Z1) and production duration, balancing the need for efficient transportation and inventory management. By selecting a warehouse closer to the construction site, the RO model reduced the likelihood of delays and improved supply chain responsiveness. The optimal production period was six weeks for initial inventory preparation and three weeks for production during the assembly phase, ensuring cost-effective operations while mitigating risks from uncertain demand conditions.

3.D.3. Study Gaps and Opportunities for Future Research

The study on risk-averse supply chain management for modular construction projects provides a comprehensive model for addressing uncertainties in logistics, but like many research endeavors, it leaves several areas unexplored. Identifying these gaps and proposing future research opportunities is critical to advancing the field and enhancing the application of robust optimization models across different industries. By addressing these gaps, future research can build on the current model's strengths while expanding its applicability and efficiency under various operational conditions.

Table 4: Gaps and Opportunities for Future Research for "Risk-averse supply chain for modular construction projects"

Identified Gaps In Study	Description	Opportunities for Future Research
Limited Focus on Upstream Supply Chain Risks	The study focuses primarily on downstream logistics uncertainties, such as transportation delays and site demand variations, without accounting for upstream risks like manufacturing delays or raw material shortages. These factors can significantly disrupt modular construction projects and addressing them could enhance supply chain resilience.	Future studies could extend the current model by integrating upstream supply chain risks, such as supplier delays or material shortages. This would provide a more holistic approach to supply chain risk management and improve overall project performance.
Narrow Industry Focus	The study's focus on modular construction limits the model's application to this specific field. Modular construction, while important, is not the only industry facing complex logistics challenges. Other sectors with large-scale, uncertain supply chains, such as healthcare, energy, or manufacturing,	The principles of risk-averse supply chain management can be applied to industries beyond modular construction. Exploring the model's application in industries such as healthcare, energy, or large-scale manufacturing would broaden its utility and demonstrate its versatility in addressing logistics uncertainties in various sectors.

	could also benefit from this type of risk-averse supply chain model.	
Advanced Risk Aversion	The study employs a general risk-aversion model but does not fully explore more complex risk metrics or decision-making approaches that could cater to different levels of risk tolerance. Understanding how varying degrees of risk aversion impact decision-making would provide more flexibility in practical applications.	Future research could explore more sophisticated risk-aversion metrics or decision-making frameworks that consider varying levels of risk tolerance. These metrics could be tailored to different stakeholders' preferences, offering more precise tools for managing risk in supply chain operations.

3.E."Integrated Framework for Project Portfolio Planning Management Under Resource Constraints" (Master's thesis, Stevens Institute of Technology)

The thesis explores the challenges of managing project portfolios effectively, focusing on maximizing returns and optimizing resource allocation. It introduces an integrated framework for project portfolio management, addressing both the planning and implementation phases. The thesis emphasizes that separating these phases often leads to suboptimal outcomes due to uncertainties during execution.

3.E.2 Study Findings

The study classified 23 papers into eight groups based on portfolio planning direction (top-down vs. bottom-up) and breadth (planning vs. implementation). Most studies focused on either planning or implementation but rarely integrated the two. Top-down methodologies were dominant, reflecting interdependencies among projects, while bottom-up approaches treated projects independently.

Integrated strategies that consider implementation uncertainties during planning yielded better portfolio performance. It was suggested that allocating idle resources during planning helps mitigate potential implementation disruptions, enhancing overall earned portfolio value.

3.E.3. Gaps and Opportunities for Future Studies

Here are a l few of the gaps and opportunities for future research.

Table 5: Gaps and Opportunities for Future Research--Integrated Framework for Project Portfolio Planning Management Under Resource Constraints

Identified Gaps In Study	Description	Opportunities for Future Research
Lack of Research on Integrating Both Planning and Implementation Phases	The primary gap identified is the lack of research on integrating both planning and implementation phases. Most studies focus only on short-term	Future research could develop more sophisticated models that incorporate both planning and implementation phases under resource constraints,

	gains, often neglecting the impact of execution uncertainties.	considering dynamic changes during execution.
Limited Exploration of Bottom-up Methodologies	Another gap is the limited exploration of bottom-up methodologies, which could provide more insights into optimizing independent project portfolios.	There's also potential to explore hybrid methodologies, combining top-down and bottom-up approaches to improve decision-making for diverse portfolio types.
Need for Practical Application and Real Time Resource Adjustments	While the proposed integrated framework shows promise in theoretical simulations, there is a lack of empirical studies that apply it in real-world project environments. The gap lies in the absence of field-tested validation, where resource adjustments and strategy changes can be dynamically observed and analyzed in real-time within actual organizations. This limits the understanding of how the framework performs under real operational pressures, diverse team dynamics, and evolving project scopes.	Emphasis should be placed on practical applications and real-time resource adjustments to further validate the proposed integrated framework.

3.F. "A clustering-based review on project portfolio optimization methods. International Transactions in Operational Research"

The paper offers a comprehensive review of project portfolio optimization (PPO) methods using a clustering-based approach to categorize existing research. It aims to clarify connections among various methodologies and identify dominant trends in project portfolio management (PPM). The review covers over 3,183 publications, narrowed down to 298 key papers, which are grouped into 24 clusters. These clusters are analyzed to highlight specific techniques, from metaheuristics and fuzzy logic to multi-criteria decision analysis (MCDA) and hybrid optimization models.

The review emphasizes that optimization methods have evolved to handle complexities such as resource constraints, uncertainty, interdependencies among projects, and multi-objective decision-making. It identifies four main approaches: fuzzy modeling, decision aids, metaheuristics, and project ranking methods, each serving different needs in portfolio optimization. The paper concludes by highlighting the potential of hybrid models, which integrate simulation, fuzzy logic, and metaheuristics, to address real-world challenges in dynamic and uncertain environments.

3.F.1. Study Results

The review indicates that metaheuristic approaches are the most scalable and adaptable for large portfolios and realistic constraints, while fuzzy modeling is more suited for smaller, uncertainty-heavy portfolios.

Simheuristics and hybrid models are the better choice for dynamic project management due to their adaptability, scalability, multi-objective capability, and robustness. They are well-suited for environments where real-time decision-making, uncertainty management, and ongoing optimization are necessary. These models, which integrate simulation, fuzzy logic, and metaheuristics, offer the best performance for dynamic, complex, and real-world scenarios.

The study concludes that decision-aid techniques (like the Analytic Hierarch Process (AHP) and the Analytic Network Process (ANP)) work well for ranking projects in simpler scenarios but are limited by scalability issues.

Efficient frontier approaches, focusing on Pareto optimality, provide decision-makers with multiple options, enabling better trade-off analysis between conflicting objectives.

3.F.2. Study Gaps and Opportunities for Future Research

The hybrid cluster, particularly simheuristics, presents a promising avenue for dynamic project management. However, addressing the identified gaps can significantly enhance their practicality, scalability, and effectiveness across diverse real-world applications. There are ample opportunities for integrating real-time data, improving computational efficiency, and expanding decision-making capabilities, paving the way for more adaptive and robust models.

3.G. Robust optimization for hurricane preparedness (International Journal of Production Economics)

The paper introduces a robust optimization framework for enhancing hurricane preparedness, focusing on minimizing social costs associated with logistics, deprivation, and fatalities. It proposes two optimization models: a single-stage adaptive robust model to determine the optimal timing for deploying resources to Points of Distribution (PODs), and a two-stage robust model that integrates recourse actions for more adaptable decisions regarding POD location, stockpile capacity, and flow. The authors emphasize the importance of accounting for deprivation costs, which quantify human suffering due to delayed relief, making this model more aligned with humanitarian logistics compared to traditional cost-minimization models.

3.G.1. Equations and Parameters

In the study, there are two models that are based on robust optimization principles with key components.

3.G.1.1. Objective Function

The objective function minimizes the total travel cost of different demands to corresponding facilities

Equation 33: Objective Function for Hierarchical Approach to Modeling Hurricane Disaster Relief Goods Distribution

Objective Function:

$$Minimize \sum_{i \in I} \sum_{j \in J} \sum_{s \in S} d_{ij} h_{is} Y_{ijs}$$

Where:

$i \in I$ represents nodes with demand.

$j \in J$ potential facility locations.
$s \in S$ represents demand levels and facility types.
d_{ij} represents the distance between levels and facility types.
h_{is} represents demand at node I for typ of s facility.
Y_{ijs} represents variable indicating if node I is served by facility j of type s.

3.G.1.2 Objective Function Constraints

The Constraints for the objective function are as follows:

3.G.1.2.1 Assignment Constraints

Equation 34: Assignment Constraints

$$\sum_{s \in S} \quad Y_{ijs} = 1, \forall i \in I, \forall s \in S$$

This constraint ensures that each demand node is served by one facility.

3.G.1.2.1 Capacity Constraints

The capacity constraint is the minimum and maximum capacity constraints for facilities to ensure efficient resource distribution.

Equation 35: Capacity Constraints

$$z_{jt} \geq b_{jt} \cdot X_{jt}, \qquad z_{jt} > B_{jt} \cdot X_{jt}$$

Where z_{jt} is the total demand served by a facility of type t, and b_{jt} and B_{jt} represents the minimum and maximum capacities.

3.G.1.2.2 Closest Assignment Constraint

Equation 36: Closest Assignment Constraint

$$Y_{ijs} = 0, if \ \ d_{ij} > D_{is}$$

This constraint limits the maximum distance from demand notes to assigned facilities.

3.G.2. Study Results and Findings

For the comparison between CM and HCM models, the HCM model placed 16 facilities in both scenarios tested (40% and 50% of the population staying behind). The HCM produced lower average travel costs for lower-level demands (Type III services) compared to the CM. In the 50% scenario, the supplies-to-person ratio improved from 1.6 (CM) to 1.3 (HCM), indicating more efficient resource allocation. While the HCM effectively minimized travel distance and differentiated demand types, it required longer computational times, limiting its immediate practicality. For the computational performance, the HCM took significantly longer to solve (up to 93.59 hours) compared to the CM, highlighting the need for faster solution methods for large-scale implementations.

3.G.3. Study Gaps and Opportunities for Future Research

Here are some gaps and opportunities for future research.

Identified Gaps	Description	Opportunities for Research
Computational Complexity	HCM requires extensive computational time.	Develop **faster algorithms** to improve scalability for large-scale implementations.
Dynamic Resource Allocation	Static assumptions for POD locations and capacities.	Explore **adaptive frameworks** to respond to real-time changes during disasters.
Incorporating Real-Time Data	Limited integration of dynamic, real-time data.	Integrate **real-time demand forecasts** for more responsive disaster planning.
Broader Applications	Focused solely on hurricane preparedness.	Apply the robust optimization framework to other disaster scenarios (e.g., earthquakes, floods).

3.H. Detailed Analysis of "A Hierarchical Approach to Modeling Hurricane Disaster Relief Goods Distribution"

The article develops a hierarchical model to optimize the distribution of relief goods following hurricanes. The authors focus on improving the siting of Points of Distribution (PODs) using a Hierarchical Capacitated-Median (HCM) model that incorporates both Geographic Information Systems (GIS) and spatial optimization techniques. The model is designed to minimize travel distance for victims while considering different types of aid facilities, which range from basic to advanced relief services. The study uses Leon County, Florida data to test the model's efficacy, comparing it to a non-hierarchical Capacitated-Median (CM) model.

3.H.1. Equations and Parameters

In the study, the HCM model extends the CM problem by including hierarchical considerations, and introducing distinct levels of relief services, each with specific capacities.

3.H.1.1. Objective Function

The objective function minimizes the total travel cost of different demands to corresponding facilities

Equation 37: Objective Function for Hierarchical Approach to Modeling Hurricane Disaster Relief Goods Distribution

Objective Function:

$$Minimize \sum_{i \in I} \sum_{j \in J} \sum_{s \in S} d_{ij} h_{is} Y_{ijs}$$

Where:

$i \in I$ represents nodes with demand.
$j \in J$ potential facility locations.
$s \in S$ represents demand levels and facility types.
d_{ij} represents the distance between levels and facility types.

h_{is} represents demand at node I for typ of s facility.
Y_{ijs} represents variable indicating if node I is served by facility j of type s.

3.H.1.2 Objective Function Constraints

The Constraints for the objective function are as follows:

3.H.1.2.1 Assignment Constraints

Equation 38: Assignment Constraints

$$\sum_{s \in S} Y_{ijs} = 1, \forall i \in I, \forall s \in S$$

This constraint ensures that each demand node is served by one facility.

3.H.1.2.2 Node Demand Constraint

Equation 39: Node Demand Constraint

$$Y_{ijs} \leq X_{jt}, \forall i \in I, \forall j \in J, \forall s \in S$$

This constraint assures that a node's demand for a service level s is served by a facility of level s or higher.

3.H.1.2.3 Facility Constraint

Equation 40: Facility Constraint

$$\sum_{s \in S} X_{ijs} = 1, \forall j \in J$$

This constrat keeps two facilities of the same or different type from being assigned to the same location, preventing facility co-location.

3.H.1.2.4 Demand Capacity Tracking

Equation 41: Demand Capacity Tracking Constraint

$$\sum_{s \in S} z_{jt} \leq \sum_{s \in S} \sum_{i \in I} h_{is} Y_{ijs}, \forall j \in J$$

This Constraint defines the capacity tracking variable z_{jt} to be the total demand of all levels servied at node j by a type-t facility.

3.H.1.2.4.1 Capacity Constraints for Capacity Tracking

The capacity constraint is the minimum and maximum capacity constraints for facilities to ensure efficient resource distribution.

Equation 42: Minimum Levels of Facility Capacity Constraint

$$z_{jt} \geq b_{jt} \cdot X_{jt}, \qquad z_{jt} > B_{jt} \cdot X_{jt}, \forall i \in I, \forall j \in J, \forall s \in S$$

Where z_{jt} is the total demand served by a facility of type t, and b_{jt} and B_{jt} represents the minimum and maximum capacities.

3.H.1.2.5 Closest Assignment Constraint

Equation 43: Closest Assignment Constraint

$$\sum_{k\in J|d_{ik}\leq d_{ij}} Y_{ijs} \leq X_{jt}, \quad \forall i \in I, \forall j \in J, \forall s \in S, \qquad \forall t \in S|t \geq s$$

This is the closest assignment constraint which mandates that the demand nodes are served by their closest sited facility.

3.H.1.2.5.1. Maximum Travel Distance from a Node to a Facility Constraint

Equation 44: Maximum Travel Distance from a Node to a Facility

$$Y_{ijs} = 0, \forall i \in I, \forall j \in J, \forall s \in S|\ d_{ij} > D_{is}$$

This constraint limits the maximum distance from demand notes to assigned facilities. This is based on medium distance from that node to all potential facilities.

3.H.1.2.5.2. Single Decision Variable Assignment Constraint

Equation 45: Single Decision Variable Assignment Constraint

$$Y_{ijs} \in \{0,1\}, X_{ijs} \in \{0,1\}, \qquad z_{jt} \geq 0, \forall i \in I, \forall j \in J, \forall s \in S, t \in S$$

This constraint defines decision variables and enforces single assignment.

3.H.2. Study Results and Findings

For the comparison between CM and HCM models, the HCM model placed 16 facilities in both scenarios tested (40% and 50% of the population staying behind). The HCM produced lower average travel costs for lower-level demands (Type III services) compared to the CM. In the 50% scenario, the supplies-to-person ratio improved from 1.6 (CM) to 1.3 (HCM), indicating more efficient resource allocation. While the HCM effectively minimized travel distance and differentiated demand types, it required longer computational times, limiting its immediate practicality. For the computational performance, the HCM took significantly longer to solve (up to 93.59 hours) compared to the CM, highlighting the need for faster solution methods for large-scale implementations.

3.H.3. Study Gaps and Opportunities for Future Research

Here are some gaps and opportunities for future research.

Table 6:"A Hierarchical Approach to Modeling Hurricane Disaster Relief Goods Distribution" Study Gaps and Opportunities in Future Research

Identified Gaps In Study	Description	Opportunities for Future Research

Improving Computational Speed	Current HCM models had long run times.	Future research should focus on developing faster algorithms or heuristics to enable real-time decision-making in disaster scenarios.
Incorporating Network Uncertainty	The current model assumes the road network remains functional after a hurricane, which is often unrealistic.	Future studies could introduce stochastic elements to account for potential disruptions, such as flooded roads or debris blockages.
Fixed Costs of Facility Types	The model does not consider differences in the setup costs of various PODs.	Integrating fixed charge terms could allow for a more comprehensive cost-benefit analysis.

3.I. Detailed Analysis of "Optimization of renewable energy project portfolio selection using hybrid (AIS-AFS) algorithm in an international case study."

Renewable energy is a hotly debated topic in every part of the world today. Renewable energy is energy that is created from natural sources and is replenished at a higher rate than they are consumed (e.g. sunlight, water, and wind). Non-renewable resources that take hundreds of millions of years to form (e.g. fossil fuels, coal, oil, gas). The negative fact about fossil fuels is that harmful greenhouse gas emissions, such as carbon dioxide, are generated when they are burned to produce energy. There is also a limited quantity of non-renewable resources. Renewable energy creates far less emissions than fossil fuels. It is assumed that transitioning from fossil fuels to renewable energy will address most climate crises. Additionally, many countries are working on becoming energy-independent and trying to build out large energy portfolios that include both renewable and non-renewable energy resources, with an emphasis on increasing their renewable energy portfolios. These portfolios must be robust and be able to support not only the energy requirements but also create a non-negative portfolio (i.e. the present value of the portfolio should be positive). One of the most important aspects of a comprehensive portfolio is having a renewable energy portfolio.

Based on this topic, I conducted a literature review on "Optimization of renewable energy project portfolio selection using hybrid (AIS-AFS) algorithm in an international case study" which discusses the optimization of renewable energy project portfolio topic in detail. Evaluating investment requires considering profitability as well as investment risk. This study uses a multi-objective mathematical model to optimize a Renewable Energy Project Portfolio (REPP). The goal is to maximize the portfolio's net present value and minimize investment risk concurrently. The study introduces a novel approach to renewable energy portfolio optimization, effectively bridging theoretical advancements with practical applications.

Ultimately, this critique will focus on three key areas: reliability and validity of methods, generalizability of findings, and the evaluation of the mathematical model, solution methods, and sensitivity analyses. This critique will also highlight strengths, limitations, and recommendations for improving the study's robustness and applicability.

3.I.1. Reliability and Validity of the Paper

3.I.1.a. Reliability

In terms of reliability, the authors of the study addressed reliability in a few ways. First, the authors developed a comprehensive dataset. The dataset was collected from 137 renewable energy projects from over 20 countries and sourced from reputable repositories such as the National Renewable Energy Laboratory (NREL), Lilliestam et al., and the World Bank. Getting the data from these reputable repositories ensured the diverse and credible inputs. Additionally, the global coverage of the dataset ensured the consistency of the wide range of investment scenarios. This also enhanced reliability.

Secondly, the authors rigorously tested and evaluated various mathematical algorithms. To optimize the objective functions of this mathematical model, a hybrid meta-heuristic algorithm combining Artificial Immune System (AIS) and Artificial Fish Swarm (AFS) algorithms was introduced. This hybrid (Hybrid Fish Swarm and Immune System Algorithm (HFSIS)) model was tested against standalone AIS and AFS methods to ensure its robustness and comparative efficiency in solving the REPP optimization problem. The proposed model was optimized using the dataset described above and considered various portfolio sizes of 3, 5, 10, and 15 various renewable energy projects. The numerical results indicated that, at a specific investment risk threshold, the proposed hybrid algorithm outperforms both AIS and AFS in terms of profitability. Additionally, assessing the geographical distribution of selected projects revealed a deliberate effort to avoid concentration in any specific region. This demonstrated the diversification of REPP by identifying optimal investment opportunities globally.

3.I.1.b. Validity

In terms of validity, I evaluated the paper based on construct validity, internal validity, and external validity.

The goal of construct validity is to ensure that the model represents real-world scenarios and addresses the research objective effectively. The overall construct validity can be seen in the robust mathematical model that incorporates the key real-world variables, such as net present value (NPV), investment risk, temporal considerations (i.e. investment/construction and exploitation periods), project lifetime integration, workforce employment (i.e. associated costs), financial flexibility (i.e. loans to enhance investment prospects in renewable projects, initial capital, funding periods throughout the projects life cycle), country-specific loan interest rates, and loan repayments. The mathematical model used in this study aligns with practical concerns for renewable energy investments, enhancing the model's relevance.

For overall internal validity, the model was validated using a small-scale test problem in GAMS software and CPLEX solver. The results were consistent and logical, which demonstrated the mathematical framework's correctness. The results obtained from these optimization calculations offer valuable insights and form a reliable basis for decision-making. Additionally, the comparison with other algorithms (e.g., AIS and AFS) further reinforces reliability. This is further proven when the HFSIS hybrid algorithm consistently outperforms in areas like profitability at specific risk thresholds.

In terms of external validity, the dataset spans many diverse economic, geographic, and technological conditions, which enhances the generalization of the model. Also, the inclusion of solar, wind, and other renewable energy projects from 20 countries and their broad representation of investment scenarios allows the findings to apply globally.

3.I.1.c. Problems Identified

There are some problems that I have identified with the reliability and validity noted above. The reliability issue that I noticed is the dependency of secondary data. The accuracy of the results depends on the integrity of external sources. For example, if the data from the World Bank or the National Renewable Energy Laboratory (NREL) contains inaccuracies, the model's reliability may be compromised. Another reliability issue has to do with a validation scale. In other words, the data set used in this article may be considered a small-scale test. These types of tests are often used for validation. However, the results may not reflect the complexity or scale of a full dataset. A real-world test on a larger dataset would provide stronger reliability.

As for other validity concerns, static assumptions could potentially be an issue. For example, static assumptions like fixed interest rates or linear depreciation can oversimplify the dynamic financial environmental conditions, which could lead to misrepresentation of project viability. Additionally, the study takes into account limited environmental factors and does not consider possible environmental-related risks (i.e. global warming, rising flood waters, drought) or adoption of environmental policies that could influence real-world applicability.

Another problematic area is the limited scope of the portfolio size. The study analyzes portfolio sizes of 3, 5, 10, and 15, but this may not properly generalize scenarios with larger or smaller portfolio values. Additionally, there is a geographical clustering that may affect validity. Although the study looks into the global coverage of projects, the selected projects are concentrated in certain areas or regions. This may result in limited insights for underrepresented areas or regions.

Additionally, while the dataset is comprehensive, its validity depends on the accuracy of the sources. Potential biases in data collection or reporting (i.e. underestimating risk or overestimating returns) could impact the results. Secondly, the HFSIS hybrid algorithm's reliance on specific assumptions (i.e. linear depreciation and fixed interest rates) might not fully capture the dynamic environmental conditions.

3.I.1.d. Why the Methods Will Produce Relevant Data

It is also important to point out why the methods will produce relevant data. First, the research questions and the multi-objective optimization model are in alignment with the research questions. The multi-objective optimization model addresses the issues of balancing maximization of NPV and minimization of risk which directly map to a research question.

Next, the advanced methodology of a hybrid model also produces relevant data. The hybrid algorithm is explicitly designed to balance the dual objectives of maximizing NPV and minimizing risk, directly addressing the research goals. Secondly, the use of the Efficient Investment Frontier (EIF) visualizations provides clear insights into the risk-return-trade-offs, enabling stakeholders to make informed decisions.

3.I.1.e. Potential Problems Identified With These Methods

There are potential problems that may arise with the methods mentioned above. The first issue has to do with algorithm dependence. The hybrid algorithm is sensitive to parameter adjustments such as step size, cloning rate, and visual radius. If the parameters do not reflect optimal values, this could affect overall reliability and model reproducibility.

Additionally, when it comes to validation and generalizability because the study is validated using GAMS software and not real-world case studies, the applicability of the study is limited.

Lastly, the static parameters (e.g. inflation rates) also makes the model less adaptable to the volatile and dynamic markets.

3.I.1.d. Recommendations for Improvements

To improve reliability, it is recommended that a real-world dataset with a more diverse list of renewable energy technologies and regions be incorporated in model testing and evaluations. This will help enhance the dataset and model output robustness and provide better insights. Additionally, along with more robust data, real-world case studies and historical investment data should be used. Overall comparisons of actual and modeled output can be compared for better analysis.

To improve validity, it is recommended that parameters reflect more variability (i.e. interest rates). Additionally, the model risks should be increased to consider other various factors (i.e. regulatory policies, technology challenges, environmental risk). This will provide a more accurate and realistic model output that fits past, current, or future situations.

Lastly, to improve generalizability, I recommend that the model should be tested using larger datasets, larger portfolio sizes, and underrepresented regions. This will improve the model's overall applicability.

If these concerns are addressed, the research can improve reliability, validity, and relevance. This will help answer research questions accurately and mitigate limitations.

3.I.2. Generalization of Findings

This section will discuss how the findings of this study are moderately generalizable. However, it is important to note that there are strengths and limitations to these applications (i.e. population diversity, region, situation).

3.I.2.a. How generalizable are the findings in the paper?

Overall, the findings of the study are moderately generalizable. There are some strengths and limitations to the applicability of the study across different populations, regions and situations. I discussed a few generalizable findings above, but there are a few others I will also discuss here.

The first strength I would like to discuss is dataset diversity. In section 3.a.Reilabity, I discuss the reliability of the dataset collected from 137 renewable energy projects from over 20 countries. This aspect is also important from a standpoint of incorporation of a wide range of economic and geographic conditions. The parameters of the data set such as country-specific inflation rates, interest rates, and workforce costs are considered, making the findings of the model output very adaptable in a financial landscape.

However, it is also important to note the limitations of these strengths. Although the dataset has specific inflation and interest rates for specific countries, these rates remain static. The model assumes a static interest rate and inflation rate. This is a limitation because it does not reflect the dynamic nature of these parameters. In the real world, these numbers fluctuate and impact investment decisions. Another static number that is a limitation in this area is the workforce numbers. These are also assumed to be static, which does not truly reflect the dynamic nature of the labor market.

Secondly, the broad energy coverage is another strength to discuss. The dataset also includes different renewable energy technologies (i.e., solar, wind) that provide a wide spectrum of renewable energy scenarios. This wide spectrum of scenarios helps ensure that findings are not skewed for a specific region.

However, the limitation to this is that the optimized portfolios select projects from specific regions (i.e., United States, Africa) and underrepresent others (i.e., South America). This highlights regions with economic and environmental advantages and limits insights for regions that are not as economically and environmentally favorable. This issue, paired with the issue of limited representation of small-scale projects, makes the findings less applicable to communities with less economic and environmental affluence. Additionally, the findings are also less applicable to community-level or small-scale renewable energy initiatives, and regions with emerging markets located in lower investment strength.

3.I.2.b. Can the findings be applied to other populations or situations?

These findings can be applied to global or multinational investors looking to optimize large-scale portfolios in well-developed renewable energy markets. However, these findings are less applicable to small or local organizations, governments, or investors who are focused on smaller or region-specific renewable energy solutions. Additionally, the study's framework is most applicable to stable economic environments where inflation and interest rates are predictable. Lastly, this study is also applicable in situations where investment decisions prioritize profitability and risk minimization over social or environmental considerations.

3.I.2.c. If not, how can one overcome such limitations?

While the study provides valuable insights for large-scale renewable energy investments, it is static assumptions, geographical clustering, and limited representation of small-scale projects constrain generalizability. These limitations can be overcome in the following ways:

a. Incorporating dynamic economic modeling by using time-varying interest and inflation rates to simulate dynamic economic conditions.
b. Expanding dataset scope by including small-scale and community-level energy projects to enhance the model's relevance to local or smaller markets.
c. Integrate policy and environmental factors (variables) representing renewable energy policies, subsidies, and environmental constraints. This will make the model more context-sensitive.
d. Validate the model using real-world case studies from different regions and investment portfolios. This will ensure that the findings hold real-world scenarios.
e. Use algorithm enhancements to compare the hybrid with alternative methods to ensure practical robustness.

By addressing these limitations through dynamic modeling, broader datasets, and integration of policy and environmental factors, the finding can be extended to a wider range of populations and situations. Additionally, these recommendations would enhance the practical applicability and impact of the research.

3.I.3. Validation of Mathematical Model, Evaluation of Solution Methods, and Sensitivity Analysis

3.I.3.a. Model Breakdown

The detailed mathematical model is a comprehensive multi-objective optimization model with two key objective functions.

The first Equation 46 is the objective function for net present value (NPV). Equation 47 is an objective function for risk minimization. These equations ensured that both profitability and risk are considered with real-world investment decision-making.

Equation 46: Objective Function for Net Present Value (NPV)

$$
\begin{aligned}
MaxZ1 = &\sum_{i=1}^{m}\sum_{t=D_i+1}^{S_i}\frac{In_{it}}{(1+r_i)^t}x_i \\
&-\sum_{i=1}^{m}\sum_{t=0}^{D_i}\frac{IO_{it}}{(1+r_i)^t}x_i \\
&-\sum_{i=1}^{m}\sum_{t=D_i+1}^{S_i}\frac{d_{it}}{(1+r_i)^t}x_i \\
&-\sum_{i=1}^{m}\sum_{t=D_i+1}^{S_i}PR2_i\left(\frac{1}{(1+r_i)^t}\right)x_i-\sum_{i=1}^{m}\sum_{t=0}^{D_i}Jn_{it}N_{it}x_i+\sum_{i=1}^{m}\frac{De_i}{(1+r_i)^{S_i}}x_i
\end{aligned}
$$

Equation 47:Objective Function For Risk Minimization

$$\mathrm{MinZ2} = \sum_{\mathrm{i}=1}^{\mathrm{m}}\mathrm{RISK_i\,x_i}$$

Subject to:

Equation 48: Constraint To Ensure That the Initial Investment is Less Than the Investor's Budget

$$\sum_{\mathrm{i}=1}^{\mathrm{m}}\{\mathrm{IO_{i0}} - \mathrm{L_i}\} \leq \mathrm{w_0}$$

Equation 48 ensures that the amount of money for the initial investment is less than the investor’s budget.

Equation 49: Constraint of Remaining Budget For Each Pay Period

$$\sum_{\mathrm{i}=1}^{\mathrm{m}}(\mathrm{In_{it}} - \mathrm{PR_{it}} + \mathrm{De_i})\mathrm{x_i} + \mathrm{w_{t-1}} = \mathrm{w_t}\forall \mathrm{t} = 1,2,\dots,\mathrm{T}$$

Equation 49 provides the remaining budget for each pay period.

Equation 50: Constraint That Makes Sure Specific Projects Have Loans

$$\mathrm{L_i} \leq \mathrm{C_b x_i}\forall \mathrm{i} = 1,2,\dots,\mathrm{m}$$

Equation 50 makes sure that loans are taken out for specific projects.

Equation 51:Constraint For Generating the Standard Amount of Energy For a Specific Project

$$\sum_{t=D_i+1}^{S_i} (Ca_{it} - SC_{it})\, x_i \geq 0 \forall i = 1,2, \dots, m$$

Equation 51 makes sure that each established project generates the standard amount of energy for that specific project.

Equation 52: Constraint to Keep the Number of Projects From Exceeding Max Number of Projects

$$\sum_{i=1}^{m} x_i \leq K$$

Equation 52 states that the number of selected projects does not exceed the maximum number of projects.

Equation 53: Constraint for Predefined Minimum and Maximum Number of Workforces for Each Project

$$Amin_i x_i \leq \sum_{t=1}^{D_i} Jn_{it}. N_{it} x_i \leq Amax_i x_i \forall i = 1,2, \dots, m$$

Equation 53 makes sure that there is a predefined minimum and maximum number of workforces for each project.

Equation 54: Decision Variable Details

$$x_i, z_i \in \{0,1\} L_i \geq 0 \forall i = 1,2, \dots, m \forall t = 0,1, \dots, D_i, \dots, S_i$$

Equation 54 explains the decision variables in more detail.

The constraints above provide a more realistic and practical feasibility for the objective functions above. These constraints take into consideration certain operational requirements that are required during real-world project management portfolio balances.

Equations 55-58 are applied to calculate PR_{it} and d_{it}.

Equation 55: Calculation for PRi

$$PR_i = L_i \frac{r(1+r)^{S_i - D_i + 1}}{(1+r)^{S_i - D_i + 1}} \forall i = 1,2, \dots, m$$

Equation 56: Calculation for PR1i

$$\mathrm{PR1_i} = \frac{\mathrm{L_i}}{\mathrm{S_i} - \mathrm{D_i} + 1} \forall \mathrm{i} = 1,2, \dots, \mathrm{m}$$

Equation 57: Calculation for PR2i

$$\mathrm{PR2_i} = \mathrm{PR_i} - \mathrm{PR1_i} \forall \mathrm{i} = 1,2, \dots, \mathrm{m}$$

Equation 58: Calculation for dit

$$\mathrm{d_{it}} = \frac{\left(\sum_{\mathrm{t'}=0}^{\mathrm{D_i}} \mathrm{IO_{it'}} - \mathrm{De_i}\right)}{\mathrm{n_i}} \forall \mathrm{i} = 1,2, \dots, \mathrm{m}, \forall \mathrm{t} = \mathrm{D_i} + 1, \dots, \mathrm{S_i}$$

Equation 49 is a non-linear equation. Equations 59 - 62 are applied to help linearize the model. M refers to a big number in these equations.

Equation 59

$$\sum_{i=1}^{m} (In_{it} + De_i) x_i - \sum_{i=1}^{m} (PPR_{it}) + w_{t-1} = w_t \forall t = 1,2, \dots, T$$

Equation 60

$$\mathrm{PPR_{it}} \leq \mathrm{Mx_i} \forall \mathrm{i} = 1,2, \dots, \mathrm{m}$$

Equation 61

$$\mathrm{PPR_{it}} \leq \mathrm{PR_{it}} + \mathrm{M}(1 - \mathrm{x_i}) \forall \mathrm{i} = 1,2, \dots, \mathrm{m}$$

Equation 62

$$\mathrm{PPR_{it}} \geq \mathrm{PR_{it}} - \mathrm{M}(1 - \mathrm{x_i}) \forall \mathrm{i} = 1,2, \dots, \mathrm{m}$$

3.I.3.b. Model Strengths

The overall strength of this model includes the following points:

- The model was validated using a small-scale test problem in GAMS with the CPLEX solver. This ensured that the mathematical framework and constraints were logically sound.
- Equations 48-54 represent constraints that provide a more realistic and practical feasibility for the objective functions above. These constraints take into consideration certain operational requirements that are required during real-world project management portfolio balances.
- The test results, such as selecting Project 1 and allocating an appropriate loan amount, align with practical expectations, demonstrating the model’s internal consistency.
- The inclusion of key variables like net present value (NPV), investment risk, workforce costs, inflation rates, and project lifetimes ensures that the model captures real-world factors affecting renewable energy investments.
- Non-linear constraints, Equation 49, were linearized using appropriate techniques (i.e., Equations 59 - 62), enhancing computational efficiency without compromising the model's accuracy.

3.I.3.b. Model Weaknesses

Although the model had some strengths, some weaknesses were also identified. This section will discuss these weaknesses.

The weaknesses of these models were as follows:

- The validation was performed on a small-scale dataset. This approach does not test the model's robustness in large-scale, real-world scenarios where complexities like interdependencies among projects or dynamic market conditions may arise.
- Fixed inflation rates (r_i), interest rates, and workforce costs reduce the model's adaptability to volatile economic conditions, compromising its applicability to markets with high financial variability.
- While the model includes financial elements like loans and depreciation (Equation 58), it assumes linear depreciation and fixed repayment schedules, which may not reflect the complexities of real-world financial structures or dynamics like variable depreciation rates or market-driven salvage values.

3.I.4. Evaluation of the Efficiency of the Proposed Methods

In this section of the study, AIS, AFS, and HFSIS algorithms were implemented and EIF was obtained using the project size to calculate REPP. The strengths and weaknesses of the evaluation of the efficiency of the proposed methods (i.e. HFSIS Hybrid, AFS, AIS algorithms) are discussed next.

3.I.4.a. Strengths

Some of the algorithm's strengths include the hybrid algorithm performance superiority and model scalability. The HFSIS hybrid algorithm consistently outperformed much better than the AFS or AIS models. The results showed that the hybrid algorithm achieved higher profitability at given risk thresholds, making it a more efficient optimization tool. Additionally, the hybrid algorithm also performed well across different portfolio sizes (K=3,5,10,15). This demonstrates its adaptability to different investment scenarios with varying portfolio sizes. An example of this is when K=10, a project is identified in South America, but no project has been selected at K=3 and K=5. From these results, it is clear that the proposed approach does not lead to a specific region of the world but instead has a broader global portfolio and suggests investing in these projects.

3.I.4.b. Weaknesses

Some of the algorithm's weaknesses include parameter sensitivity, lack of comparison with state-of-the-art algorithms, and computational resource dependency.

When it comes to the hybrid model, the algorithm relies on several parameters (i.e., cloning rate, visual radius, maximum iteration). Suboptimal tuning of these parameters can significantly affect the algorithm's performance. This highlights a need for more robust parameter optimization techniques.

Another identified weakness of the algorithm is that the study does not compare the hybrid model with other advanced optimization techniques. This limits the assessment of the algorithm's true efficiency.

Lastly, the study uses MATLAB R2019a software, the scalability of the hybrid algorithm to large-scale data sets with more complex constraints may increase computational demands, which is not fully addressed in the article and will require future research.

3.I.5. Critique of Sensitivity Analyses

The study performs a sensitivity analysis on the effect of parameters such as initial investment (IO) and interest rate (r_i) on NPV. The values of the parameters were fluctuated between -30% and 30%. For each fluctuation, the NPV was reported.

The IO parameters were generated from a uniform distribution with a lower limit of 20 and an upper limit of 60. The results showed that the slope of NPV decreased sharply from 20 to 30%. This showed that the initial investment is effective in calculating depreciation. By increasing the amount of the initial investment, this leads to an increase in the deprecation cost. As a result of a depreciation of costs, the costs in NPV are also reduced. Table 16 in the study, shows the results of the IO parameter sensitivity analysis. Here we can see that a 30% reduction in the IO parameter makes the NPV positive for all projects. It also shows a non-linear slope in the graph, direct and indirect influences, highlighting the different impacts of IO on NPV.

A similar relationship is shown in Table 17 and Figure 20 of the report. This table and graph show a non-linear relationship between the NPV and interest rates. The results show a the highest reduction percentage during a transition of interest rates between -20% to -10%. The analysis of NPV in relation to interest rates shows that interest rates strongly influence the value of NPV. High interest rates can lead to a negative NPV. The study finds that before a decision is made, the proper sensitivity analysis is done on the impact of an interest rate on the overall NPV of a project. Bottomline, the slightest change in interest rates can impact NPV negatively or positively.

Subsequently, these findings show the need for further research and examination with real world scenarios and examination of sensitivity analysis of other parameters.

3.I.5.a. Strengths of Sensitivity Analysis

The strengths of the sensitivity analysis are as follows:

- The sensitivity analysis was conducted on critical parameters like initial investments (IO) and interest rates. The findings show that these parameters positively or negatively impact the behavior of NPV.
- The sensitivity analysis also revealed a non-linear relationship between parameters and NPV. This shows that NPV does show realistic financial dynamics.

3.I.5.b. Weaknesses of Sensitivity Analysis

The weaknesses of the sensitivity analysis include the following:

- The sensitivity analysis only examined IO and r. Other parameters such as workforce costs and salvage values, were not analyzed. This is a missed opportunity to evaluate the model's sensitivity entirely.
- A multi-factor sensitivity analysis was also not performed. This type of analysis could have provided further insights into parameters and their behavior and impact on the model.
- The sensitivity analysis uses a uniform percentage change (-30% to +30%) for parameter variations. These parameters are not realistic in real-world scenarios and do not follow a linear pattern.

- Although sensitivity analysis assesses NPV, it does not show how parameter changes affect project selection or portfolio makeup. This is a missed opportunity and a crucial point of the study.

3.I.6. Recommendations for Improvement

Overall, the mathematical model effectively integrates profitability and risk considerations, but its validation on small-scale data and reliance on static assumptions limit its real-world applicability. The HFSIS hybrid algorithm demonstrates superior performance but requires further benchmarking. While sensitivity analyses provide valuable insights, the scope is narrow. Addressing these limitations can significantly enhance the study's robustness, practical relevance, and contribution to renewable energy portfolio optimization. Below are a few recommendations for improvement.

3.I.6.a. Enhancing Model Validation

To enhance the model validation, it is recommended to validate the model on a larger, real-world dataset to test the robustness and scalability. Additionally, it is recommended to make time-varying interest rate and inflation rates dynamic variables to reflect real-world market conditions.

3.I.6.b. Improving Solution Methodology and Efficiency

To improve the solution methodology and efficiency it would be good to use parameter tuning methods to optimize the algorithm parameters. Machine learning is an example of a parameter tuning method that can be used.

3.I.6.c. Expanding Sensitivity Analyses

As discussed earlier, it would be good to conduct a multi-factor sensitivity analysis to understand interactions between key parameters. It would also be good to evaluate the impact of these parameters and their variations on things like project selection and not just NPV. This would give a good understanding of how other parameters affect portfolio optimization as a whole.

3.I.6.d. Exploring Computational Feasibility

Exploring computational feasibility is another area to look at. This would help assess computational efficiencies on the different models, especially the hybrid algorithm. To do this, larger and more complex datasets with complex scenarios should be used. This will show the practical usability and computational feasibility of the tested algorithm.

3.I.7. Future Research Opportunities

Areas of future research opportunities for this project are explained in this section.

First, future studies should explore incorporating dynamic variables such as interest rates, inflation, and workforce costs to better reflect the real-world scenarios and markets. Additionally, this study or other studies should consider the geographical factors and regulatory risks specific to renewable energy projects within the region. Applying the model to a broader range of geographical and economic contexts could further validate its generalizability.

The other area that should be further researched is expanding the scope of the sensitivity analysis. A multi-factor sensitivity analysis was also not performed. This type of analysis could have provided further insights into parameters and their behavior and impact on the model. Although sensitivity analysis assesses NPV, it does not show how parameter changes affect project selection or portfolio makeup.

4. Literature Review of Similarities

The literature review included a diverse set of studies, each contributing unique insights into managing project portfolios under uncertainty and resource constraints. Several key similarities emerged across these methodologies.

4.1 Reliance on Optimization Techniques

Across journals and articles, optimization techniques like multistage stochastic programming and risk-averse approaches were consistently used to address uncertainties in project scheduling, resource allocation, and supply chain logistics. Additionally, many of these studies highlight multistage stochastic programming for its effectiveness in construction and healthcare settings, where rapid changes in demand are common. The authors emphasized how scenario-based decision-making helps in anticipating future outcomes, allowing for better resource distribution [1].

The journal titled *Risk-averse Supply Chain for Modular Construction* demonstrated similar principles, applying risk-averse decision-making to manage unpredictable supply chains, further supporting the adaptability of these models in volatile environments [4].

These techniques provided practical solutions for managing dynamic risks, showcasing their broad applicability in industries like project selection, healthcare, and modular construction, where unpredictability is inherent. Notably, the emphasis on scenario-based methods allows decision-makers to anticipate a range of possible outcomes and adjust plans proactively, aligning with the findings of *Risk-Averse Optimization and Control*, which proposes adaptive strategies for real-time resource adjustments [7].

4.2 Multi-Skilled Human Resource Management in Stochastic Scheduling

Several studies, particularly *Stochastic Project Management: Multiple Projects with Multi-Skilled Human Resources*, emphasized the importance of effectively managing human resources across multiple projects [2]. The models discussed in this thesis focused on integrating internal and external resources, especially under uncertain work package processing times [2]. This integration was noted to improve scheduling efficiency and resource utilization, a critical factor in sectors like construction and healthcare where fluctuating demand is common [2].

Scenario reduction techniques were also widely discussed, with a focus on enhancing computational efficiency [2]. The literature noted that as projects scale up, solving large-scale optimization problems requires effective scenario reduction to maintain manageable computation times [2]. This need for efficiency was observed in healthcare, modular construction projects and disaster response models, where time-sensitive decision-making is crucial [4].

4.3 Focus on Scenario-Based Decision-Making

All studies emphasized scenario-based decision-making as a critical similarity, particularly for better anticipating future outcomes and planning accordingly. In the *A Hierarchical Approach to Modeling Hurricane Disaster Relief Goods Distribution* article, scenario-based models were used to evaluate resource placement needs for potential disaster scenarios, allowing for adaptive responses to unforeseen conditions [8]. This aligns with project management models that aim to proactively adjust to changes in project requirements, particularly when dealing with uncertain timelines or supply chain constraints. This journal also reinforced that adaptive decision-making based on scenarios improves resource allocation efficiency, supporting real-time adjustments across sectors like disaster response, construction, and logistics.

4.4 Portfolio Management

The study *Optimization of renewable energy project portfolio selection, using hybrid (AIS-AFS) algorithm in an international case study* adds substantial depth to the analysis of optimization techniques by emphasizing hybrid algorithms (AIS-AFS) for renewable energy portfolio management [12]. This aligns with other studies in terms of leveraging optimization techniques, scenario-based decision-making, and the emphasis on computational methodologies. Additionally, there are shared themes such as multi-objective optimization (e.g., maximizing profitability and minimizing risks in uncertain environments (e.g., modular construction supply chains or budget allocation in health care projects). For example, in section 3.I., the hybrid model shares similarities with the multistage stochastic programming model in section 3.A. Both leverage advanced computational methods to address multi-objective problems under uncertainty. Additionally, like the risk-averse budget allocation model in Section 3.C., the AIS-AFS model incorporates risk minimization as a core objective in the context of renewable energy portfolio management [12].

5. Literature Review of Differences

The literature review revealed key differences in the methodologies used across the reviewed journals, articles, and theses. These distinctions encompass adaptability to real-time changes, approaches to risk management, resource deployment strategies, handling of non-financial risks, and computational efficiency. Below is a breakdown of these differences.

5.1. Adaptability to Real-Time Changes

5.1.1. Disaster Response Models

Disaster response models require immediate resource reallocation to address rapidly evolving situations, such as changing weather patterns or sudden infrastructure damage. These models emphasize real-time adaptability, as rapid adjustments are essential for effective disaster management. For example, the *Robust Optimization for Hurricane Preparedness* journal highlights that such models rely on dynamic inputs and continuously updated information, allowing for quick decision-making to ensure safety and minimize damage [8].

5.1.2. Traditional Project Management Models

In contrast, traditional project management models often operate on periodic updates, focusing more on structured adjustments rather than real-time shifts. These models, such as those discussed in *Multistage Stochastic Programming Approach in Project Selection and Scheduling*, are less equipped to handle immediate changes, which can be a drawback in projects that face unexpected delays or sudden changes in scope. For example, the current model assumes that resources (budget, staff) are available at specific points in time and do not change dynamically during the project. However, real-world projects often face time-dependent variations in resource availability, such as sudden budget cuts or shifts in personnel. Additionally, the current model does not account for the possibility of canceling or delaying projects due to unfavorable scenarios. In practice, organizations often need the flexibility to cancel projects or pause tasks when resources are severely constrained.

5.1.3 Modular Construction Projects - Stochastic Optimization in Modular Construction

Models in modular construction fall somewhere in between. While they allow for moderate adaptability, they still lack the urgency of disaster response models. Adaptations are generally made based on planned updates rather than in-the-moment changes, limiting their responsiveness to sudden market fluctuations or supply chain disruptions.

5.2. Risk Management Approaches

5.2.1 Risk-Neutral Models - Static vs. Dynamic Project Risk Models

Many project management models in this category operate under risk-neutral assumptions, focusing primarily on financial risks like cost overruns or budget adjustments. While effective in stable conditions, these models often struggle in high-stakes projects where broader, dynamic risks exist. This narrow focus limits the model's ability to respond to unforeseen events, such as regulatory changes or supply chain interruptions.

5.2.2. Risk-Averse Models - Risk-Averse Optimization and Control

In contrast, risk-averse models, such as those outlined in *Risk-Averse Optimization and Control*, focus on incorporating broader risk factors, treating uncertainties as dynamic inputs. These models emphasize proactive decision-making, which includes managing non-financial risks like workforce fluctuations or changes in stakeholder demands. This approach is particularly beneficial in disaster response and large-scale infrastructure projects, where dynamic decision-making is critical to maintaining project stability [7].

5.2.3. Healthcare and Construction Sectors

Models in healthcare and modular construction also show a mix of risk management strategies. While primarily operating with risk-neutral assumptions, they are beginning to integrate elements of risk aversion, particularly when dealing with unpredictable supply chains or patient care demands. This hybrid approach allows for some degree of flexibility but falls short of the full adaptability seen in disaster response models.

5.3. Resource Deployment Strategies

5.3.1. Broad Initial Deployment -- Adaptive Resource Deployment in Disaster Scenarios

Disaster response models often begin with a broad deployment of resources to ensure readiness for immediate action. Adaptive resource deployment in disasters is an approach that prioritizes availability and rapid mobilization, as the consequences of delayed resource deployment can be catastrophic (i.e. loss of life).

5.3.2. Minimal Initial Deployment -- Dynamic Resource Allocation under Uncertainty

In traditional project management, models often favor starting with minimal resources and scaling up based on observed needs. This strategy aims to maximize cost efficiency and minimize risks early in the project, making it well-suited for environments where immediate resource availability is not critical.

5.4. Handling of Non-Financial Risks

5.4.1 Traditional Models -- Static vs. Dynamic Project Risk Models

Traditional project management models primarily focus on financial risks, often overlooking non-financial factors such as employee turnover, stakeholder conflicts, or regulatory shifts. This limitation was highlighted in many of the articles, where models predominantly addressed cost-related uncertainties, leaving gaps in addressing broader risks that can impact project outcomes.

5.4.2. Sector-Specific Models -- Stochastic Optimization in Modular Construction

Models in healthcare and modular construction sectors have started incorporating non-financial risk factors, particularly in managing multi-skilled human resources. This approach helps address uncertainties in work package processing times, improving scheduling efficiency under fluctuating demand. However, these models still face limitations in handling broader uncertainties, such as regulatory changes or shifts in market conditions.

5.5. Computational Efficiency and Scenario Reduction

5.5.1. Large-Scale Optimization Problems -- Stochastic Optimization in Modular Construction

Models focused on large-scale projects, such as those in modular construction, emphasize the importance of scenario-reduction techniques to maintain computational efficiency. The reviewed thesis on stochastic optimization highlighted that effective scenario reduction is essential for managing the complexity of large datasets, ensuring that the models remain practical for real-world applications.

5.5.2. Sector-Specific Efficiency --Dynamic Resource Allocation under Uncertainty)

Project management models aim for long-term computational efficiency. While they benefit from scenario reduction techniques, the focus is more on optimizing resource allocation over time rather than achieving immediate results. This approach differs significantly from disaster management, where speed and adaptability take precedence.

A good example of this is in section 3.I for *Optimization of renewable energy project portfolio selection, using hybrid (AIS-AFS) algorithm in an international case study.* Unlike the modular construction models in Section 3.D., which primarily focus on logistics and supply chain risks, the AIS-AFS algorithm in Section 3.I. targets the optimization of renewable energy portfolios by integrating both financial and environmental considerations. Furthermore, while the healthcare budget allocation model (Section 3.C.) emphasizes worst-case scenarios, the AIS-AFS algorithm balances risk minimization with maximizing net present value across diverse global projects [12].

6. Future Research Focus

My future research will focus on risk-averse project management for advancing project selection and scheduling methods under uncertain conditions, especially through risk-averse modeling. This research will be a continuation of the first journal article reviewed, *A Multistage Stochastic Programming Approach in Project Selection and Scheduling*. My future research should aim to further enhance these models by exploring dynamic, multistage stochastic programming approaches that are adaptable to changing conditions.

A key area of focus will be the development of time-consistent risk measures that allow for iterative decision-making across multiple stages. This dynamic approach ensures that risk evaluations are continuously updated based on new information, enabling decision-makers to respond to both predictable and unforeseen risks effectively. Incorporating coherent risk measures into multistage models can address low-probability but high-impact events, ensuring that projects are resilient against unexpected disruptions.

Additionally, other areas that my research will investigate exploring in this article is integrating time factors into the models, potentially through Markov risk-averse frameworks. This inclusion can provide a more comprehensive view of project portfolios by balancing time-to-completion with profit maximization. Expanding these models to accommodate larger portfolios of projects, while maintaining risk aversion could offer valuable insights into managing resources effectively in dynamic, uncertain environments.

By focusing on these enhancements, future research can contribute to more adaptive, resilient project management strategies, ultimately improving decision-making under uncertainty and maximizing project outcomes across various sectors.

7. Conclusion

This literature review reveals significant advancements in project portfolio management methodologies, particularly in addressing uncertainties and resource constraints. Techniques such as multistage stochastic programming and risk-averse approaches offer promising solutions across various industries, enabling decision-makers to handle complex project environments more effectively. However, several gaps remain, including the need for more comprehensive risk models, time-dependent resource allocation, and the integration of dynamic, real-time decision-making.

Future research, as outlined in this review, will focus on enhancing project selection and scheduling methods under uncertainty through risk-averse modeling. Building on the foundations of the first journal article reviewed, *A Multistage Stochastic Programming Approach in Project Selection and Scheduling*, this research aims to develop dynamic, multistage stochastic programming models that are adaptable to changing conditions. A key emphasis will be on time-consistent risk measures that support iterative decision-making, continuously updating risk evaluations based on new information. This approach is crucial for addressing both predictable and unforeseen risks, ensuring project resilience against unexpected disruptions.

Additionally, integrating time factors into models, potentially through Markov risk-averse frameworks, will offer a more comprehensive view of project portfolios by balancing time-to-completion with profit maximization. Expanding these models to accommodate larger project portfolios while maintaining risk aversion could provide valuable insights into managing resources effectively in dynamic, uncertain environments.

By advancing these methodologies, future research will contribute to the development of more adaptive, resilient project management strategies, improving decision-making under uncertainty and maximizing project outcomes across various sectors. These advancements will enhance not only the robustness of project management practices but also their applicability to diverse, real-world scenarios, ultimately driving better project outcomes.

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